

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1707

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4

Total:

Marks:40

Annexure A

5

Code of Conduct:

I P.Kalyani(192472199) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Kalyani

ANNEXURE A

INDUSTRY PROBLEM TASK

Industry Scenario: A healthcare company has collected patient records containing age, blood pressure, cholesterol, glucose level, BMI and family history to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions such as Diet, Exercise, Medication and Monitor based on patient state transitions over time.

Task 1: Data Preparation & Inductive Learning

Data Preparation & Inductive Learning

Features → preprocessing → balanced training data → test-ready dataset



Figure 1: Task 1: Data Preparation & Inductive Learning workflow

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

Answer: Inductive learning learns a general predictive rule from labelled examples and applies that rule to unseen patient records. Here, the target variable is diabetes status, represented as Diabetic/Non-Diabetic. Suitable features include glucose level, BMI, blood pressure, age, cholesterol and family history. Glucose is a directly

relevant metabolic measurement; BMI provides information about body composition; blood pressure and cholesterol provide additional health information; age captures age-related risk patterns; and family history represents familial risk. The workflow is labelled data → preprocessing → training → validation/testing → prediction. Data leakage must be avoided, and only information legitimately available at prediction time should be used.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases). Suggest and justify a suitable balancing strategy (SMOTE / undersampling / class weights).

Answer: Because non-diabetic cases are more numerous, an unbalanced classifier may favour the majority class and miss diabetic cases. A suitable first strategy is class weighting, because it increases the importance of the minority diabetic class without discarding real observations or creating synthetic patient records. SMOTE is another option, but it must be applied only to the training data after the train-test split to avoid leakage. Under-sampling is simple but removes majority-class observations. For this scenario, class weights are a strong choice because the dataset is medical and preserving real observations is valuable. Evaluation should emphasize recall, precision, F1-score and AUC-ROC rather than accuracy alone.

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Answer: Use 70% of the data for training and 30% for testing. A stratified split should preserve the diabetic/non-diabetic class proportions in both subsets. Preprocessing should include missing-value handling, categorical encoding, and numerical scaling where appropriate. Standardisation is useful for Logistic Regression and other scale-sensitive models, while Decision Trees generally do not require feature scaling for their split decisions. Preprocessing parameters must be learned from the training data and then applied to the test data. The 70:30 split provides substantial training data while retaining a meaningful unseen test set for final evaluation.

Task 2: Decision Tree for Diagnosis

Decision Tree Diagnosis

Gini / Information Gain → interpretable classification → pruning



Figure 2: Task 2: Decision Tree for Diagnosis workflow

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

Answer: Train a Decision Tree using the preprocessed patient features and diabetes status as the target. The root node is selected using the best impurity reduction. For Gini Index, $Gini = 1 - \sum(p_i^2)$, and the preferred split has the lowest weighted impurity. For Information Gain, the preferred split produces the greatest reduction in entropy. The first three levels should show the root feature and threshold followed by its child splits and the resulting decision branches. The exact root feature and numerical Gini/Information Gain values cannot be truthfully stated from the assessment file because the actual patient dataset and calculated model output are not supplied. They should be inserted from the executed model.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

Answer: Evaluate the tree on the held-out 30% test set. Accuracy = $(TP+TN)/(TP+TN+FP+FN)$. Precision = $TP/(TP+FP)$. Recall = $TP/(TP+FN)$. F1 =

$2(\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$. False Negatives are actual diabetic cases predicted as non-diabetic. The exact numerical metrics require the actual dataset and predictions, which are not included in the source. To reduce false negatives, class weighting, threshold tuning where applicable, and recall-focused model selection can be used. Clinically, reducing missed cases can be important because an undetected condition may delay further assessment. The model should support, not replace, qualified clinical evaluation and confirmatory testing.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs postpruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Answer: Post-pruning removes branches that add insufficient predictive value. Cost-complexity pruning uses a parameter such as `ccp_alpha`; candidate values can be evaluated using validation or cross-validation. Pre-pruning restricts growth during training using limits such as maximum depth or minimum samples per leaf. Compare both models on the same held-out test set. The exact accuracy values cannot be produced from the source because no dataset/model output is supplied. For clinical use, a simpler, well-validated pruned tree can be preferable when it maintains acceptable predictive performance because it is easier to inspect and audit. However, recall, calibration, subgroup performance and the consequences of false negatives must also be considered

Task 3: Statistical Learning for Risk Stratification

Statistical Learning & Risk Stratification

Fair comparison with the Decision Tree using the same test data



Figure 3: Task 3: Statistical Learning for Risk Stratification workflow

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

Answer: Logistic Regression is suitable for binary diabetes classification because it estimates the probability of the positive class. Its probability can be written as $P(y=1|x) = 1/(1+e^{-(\beta_0+\beta_1x_1+\dots+\beta_nx_n)})$. Train it using the same 70:30 split and compare Accuracy and AUC-ROC against the Decision Tree on the same test set. The exact values cannot be calculated without the patient dataset and predictions. A statistical model can be preferable when transparent probability estimates and a relatively simple relationship between predictors and outcome are important. Decision Trees can capture nonlinear threshold relationships more naturally, so model selection should consider both predictive performance and interpretability.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Answer: For the Decision Tree, feature importance can be obtained from the total impurity reduction contributed by each feature. For Logistic Regression, feature influence can be examined through the fitted coefficients, with standardisation helping make coefficient magnitudes more comparable for numerical variables. The final answer should present the top three features from each model in a table and state whether they overlap. The exact top-three predictors cannot be named from the assessment file alone because importance values are not supplied. Agreement can occur when both models find the same variables consistently informative. Disagreement can occur because trees capture nonlinear thresholds and interactions, while Logistic Regression models relationships in the log-odds unless additional transformations or interactions are introduced.

Task 4: Reinforcement Learning for Treatment Recommendation

MDP + Q-Learning Treatment Recommendation

Diet • Exercise • Medication • Monitor → state transitions → policy



Figure 4: Task 4: Reinforcement Learning for Treatment Recommendation workflow

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

Answer: The treatment recommendation problem can be represented as a Markov Decision Process. States represent patient health stages, for example Stable/Low Risk, Moderate Risk, High Risk, Improving and Deteriorating. Actions are Diet, Exercise, Medication and Monitor as specified in the task. The reward represents health improvement, with positive values for desirable transitions and negative values for deterioration or unsafe outcomes. Transition dynamics $P(s'|s,a)$ describe how an action can move the patient from one state to another. In a classroom simulation, these transitions can be represented with a predefined transition table. For real clinical deployment, transition probabilities and reward definitions require clinical evidence, validation and governance rather than being invented for the model.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Qtable update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

Answer: Q-Learning updates the value of a state-action pair using $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$. Here α is the learning rate, γ is the discount factor, r is the reward, and s' is the next state. Across at least five episodes, the agent observes a state, selects an action, receives a reward, observes the next state and updates the Q-table. For the required two iterations, show numerical updates for at least three pairs such as $Q(s1, \text{Diet})$, $Q(s2, \text{Exercise})$ and $Q(s3, \text{Monitor})$. The exact numerical values require the transition table, rewards, α , γ and initial Q-table, none of which are supplied in the assessment document. A suitable convergence criterion is $\max |Q_{\text{new}} - Q_{\text{old}}| < \epsilon$ for several consecutive episodes, or an unchanged policy for several episodes.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

Answer: The final policy is $\pi(s) = \arg\max_a Q(s,a)$, meaning the action with the highest learned Q-value is selected for each state. The final policy should be

displayed as a state-to-action table using the trained Q-table. Exact actions cannot be determined from the source because no completed Q-table is supplied. Ethical deployment requires patient safety controls, human clinical oversight, bias evaluation across relevant patient groups, and explainability of recommendations. Medication-related actions are especially sensitive and should not be autonomously prescribed by an unvalidated classroom RL system. Audit logs, override mechanisms, privacy protection, model monitoring and validation are important safeguards. The RL system should be treated as a decision-support research prototype unless it has undergone appropriate clinical validation and governance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation